RISK PORTFOLIO REPORTING

Automated risk portfolio reporting

Muster-Sparkasse West (fiktiv)

Reporting date 2026-09-30 | Peer group PG_URBAN

Synthetic demonstration report

This report combines a shared ingestion and data-quality platform with two analytical modules: Rating & Scoring and Loss Estimation. It is a runnable prototype built exclusively on fictional Sparkassen, portfolios and cashflows.

Institution scope	**Muster-Sparkasse West (fiktiv)**	Community view or single fictional institution.
Current EAD	**799,603,375 EUR**	Positive exposure at default in the reporting snapshot.
Exposure-weighted PD 12M	**2.77%**	Configured demonstration PD mapping, not a validated model.
Observed 12M default rate	**10.34%**	Backward-looking 12-month cohort.
Observed 12M recovery rate	**17.07%**	Realised recovery cashflows through the reporting date.
Observed 12M loss rate	**82.93%**	EAD at default minus realised recoveries.
Data quality score	**99.82%**	Valid records / received records across completed batches.
Data version	**RUN-20261111005418-7B456F2D**	Latest pipeline run identifier.

SYNTHETIC DEMONSTRATION DATA - NOT REGULATORY REPORTING AND NOT REAL SPARKASSEN OR CUSTOMER INFORMATION

No real Sparkasse, customer, employee or portfolio is represented. Values are generated with a fixed seed and controlled data-quality defects for demonstration. The report is not regulatory reporting and must not be used for supervisory, accounting or customer decisions.

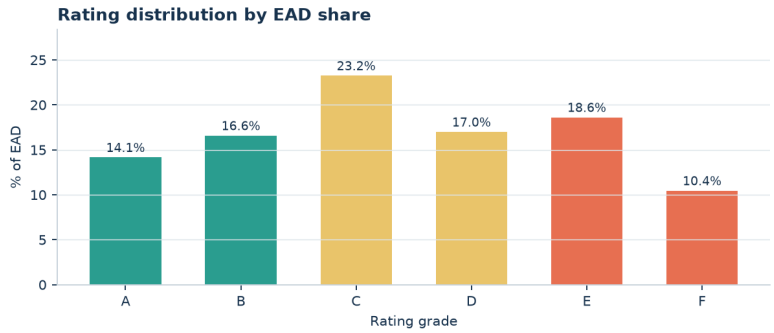
1. Executive overview

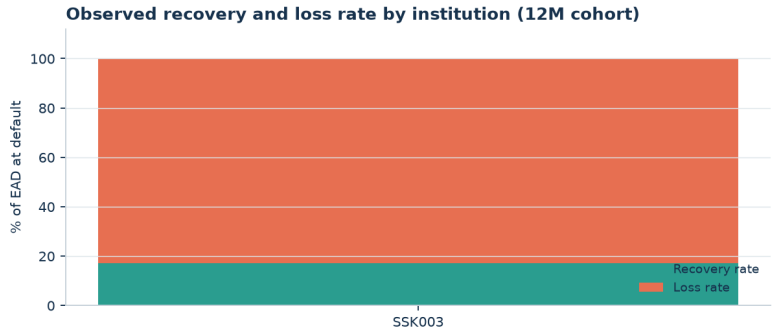
One reporting date, two analytical modules and a shared data foundation.

At 2026-09-30, the selected scope contains 557 exposures with total EAD of EUR 799,603,375. The exposure-weighted configured PD 12M is 2.77%. The observed 12-month default rate for the backward cohort is 10.34% (46 defaults on a base of 445 performing exposures). Observed recovery through the reporting date is 17.07%, giving an observed loss rate of 82.93% on EAD at default of EUR 81,690,776. Collateral coverage of current EAD is 22.46%; the secured EAD share is 22.45%.

Data quality is 99.82% record pass rate across 4 completed batches. 19 error findings and 1 warning findings were recorded; 17 records are retained in quarantine and were not loaded to curated tables. Warnings remain in curated data with their rule evidence, rather than being silently deleted.

Portfolio PD 12M	**2.77%**	Exposure-weighted configured PD.
Observed 12M default rate	**10.34%**	Count-based backward cohort.
Investment-grade share	**30.72%**	EAD rated A or B.
Upgrade / downgrade rate	**7.38% / 13.45%**	Matched 12-month migration.
Recovery rate	**17.07%**	Recoveries / EAD at default.
Loss rate	**82.93%**	Loss / EAD at default.
Collateral coverage	**22.46%**	Haircut-adjusted collateral / current EAD.
Quality score	**99.82%**	Valid / received records.



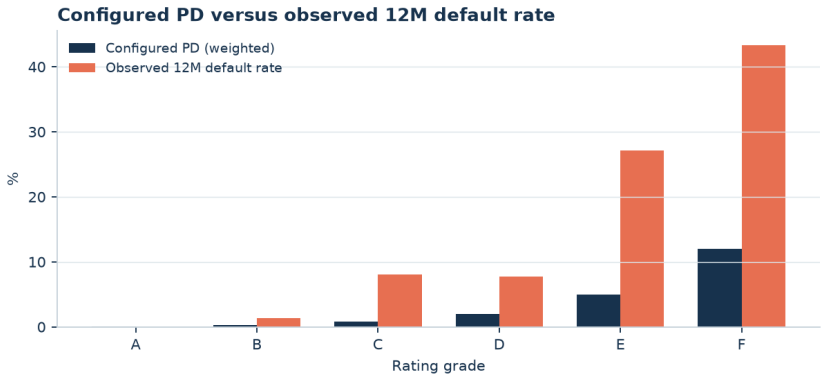


2. Rating & Scoring

Distribution, configured PD 12M, observed default rates and rating migration.

2.1 Rating distribution

Rating	Exposures	EAD (EUR)	Share count	Share EAD	PD 12M	Expected default EAD
A	83	113,123,870	14.90%	14.15%	0.10%	113,124
B	73	132,547,232	13.11%	16.58%	0.30%	397,642
C	128	185,848,403	22.98%	23.24%	0.80%	1,486,787
D	119	135,759,697	21.36%	16.98%	2.00%	2,715,194
E	96	148,782,601	17.24%	18.61%	5.00%	7,439,130
F	58	83,541,571	10.41%	10.45%	12.00%	10,024,989



2.2 Observed default rates

The observed default rate uses a 12-month backward cohort. The denominator is the set of performing exposures in the latest quarter snapshot on or before reporting date minus 12 months. A default is counted once per exposure when the first observed default date lies after that observation start and on or before the reporting date. The configured PD is shown separately and is not calibrated to the observed rate in this prototype.

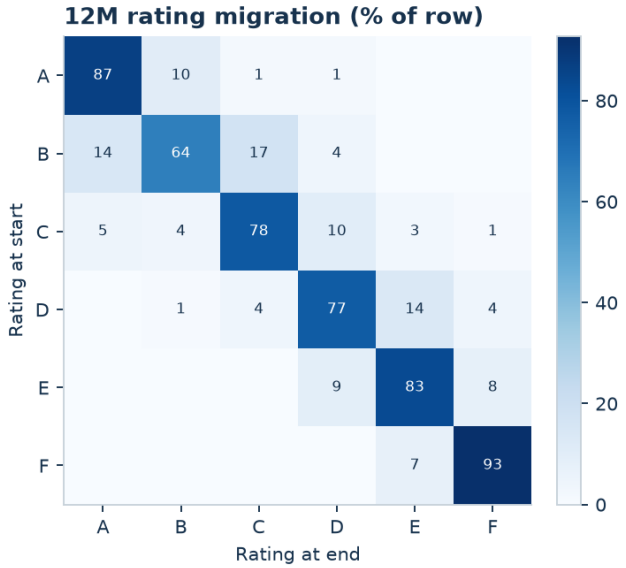
Rating	Base exposures	Base EAD	Defaults	Default EAD	Observed rate	Observed rate (EAD)	Configured PD
A	76	119,683,426	0	0	0.00%	0.00%	0.10%
B	77	112,468,660	1	371,040	1.30%	0.33%	0.30%
C	112	173,239,848	9	11,911,619	8.04%	6.88%	0.80%
D	91	92,888,311	7	5,793,632	7.69%	6.24%	2.00%
E	59	106,770,836	16	48,189,898	27.12%	45.13%	5.00%
F	30	36,795,423	13	9,753,899	43.33%	26.51%	12.00%

Bank	Base exposures	Defaults	Observed rate	Observed EAD rate	Configured PD
SSK003	445	46	10.34%	11.84%	2.10%

2.3 Rating migration

Migration compares 2025-09-30 with 2026-09-30. 461 exposures are matched. 97 entries and 36 exits are reported separately. Upgrade rate is 7.38%; downgrade rate is 13.45%.

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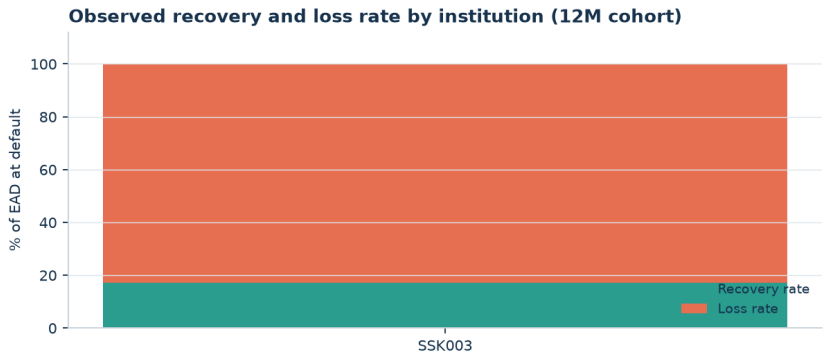
From / to	A	B	C	D	E	F
	86.76%	10.29%	1.47%	1.47%	0.00%	0.00%
	14.29%	64.29%	17.14%	4.29%	0.00%	0.00%
	4.59%	3.67%	77.98%	10.09%	2.75%	0.92%
	0.00%	1.04%	4.17%	77.08%	13.54%	4.17%
	0.00%	0.00%	0.00%	9.09%	83.12%	7.79%
	0.00%	0.00%	0.00%	0.00%	7.32%	92.68%

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3. Loss estimation

Observed recovery, loss severity, collateral coverage and cashflow timing.

The observed default cohort covers 50 defaults with EAD at default of EUR 81,690,776. The observed loss amount is EUR 67,745,412. Collateral is used only for coverage analysis and is not subtracted again from loss, preventing double counting. Observed recovery rate is 17.07%; observed loss rate is 82.93%. 28.00% of defaults have no observed recovery by the reporting date.

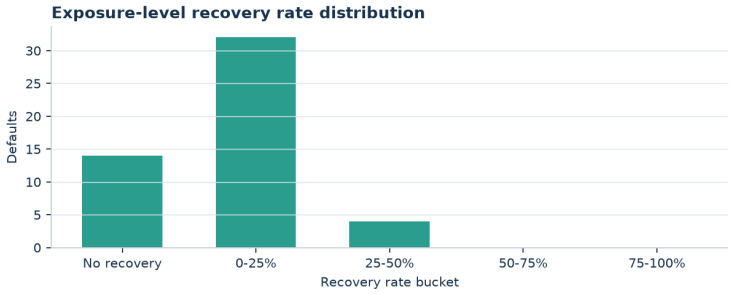


3.1 Recovery and loss by institution

Bank	Defaults	EAD at default	Recoveries	Recovery rate	Loss	Loss rate
SSK003	50	81,690,776	13,945,364	17.07%	67,745,412	82.93%

3.2 Recovery rate by rating at default

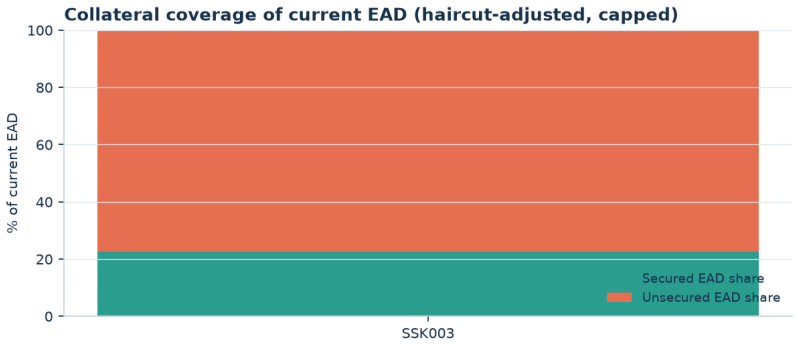
Rating	Defaults	EAD at default	Recoveries	Recovery rate	Loss rate
B	1	371,040	67,207	18.11%	81.89%
C	8	11,737,615	2,268,313	19.33%	80.67%
D	7	5,841,593	350,115	5.99%	94.01%
E	17	33,262,876	4,395,942	13.22%	86.78%
F	17	30,477,653	6,863,787	22.52%	77.48%



4. Collateral and recovery cashflows

Latest valuations, haircuts, coverage, cashflow timing and cumulative recovery.

Collateral coverage is based on the latest valuation on or before the reporting date. Configured haircuts are 20% for real estate, 5% for financial collateral and 40% for other collateral. Total adjusted collateral is EUR 179,580,660; the unsecured EAD amount is EUR 620,062,557. 0 latest collateral items exceed the 3-year valuation age warning threshold.



4.1 Collateral by institution

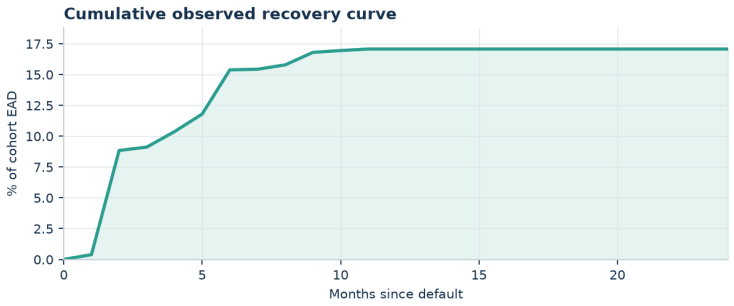
Bank	Exposures	Current EAD	Collateral value	Adjusted value	Secured EAD share	Coverage
SSK003	557	799,603,375	n/a	179,580,660	22.45%	22.46%

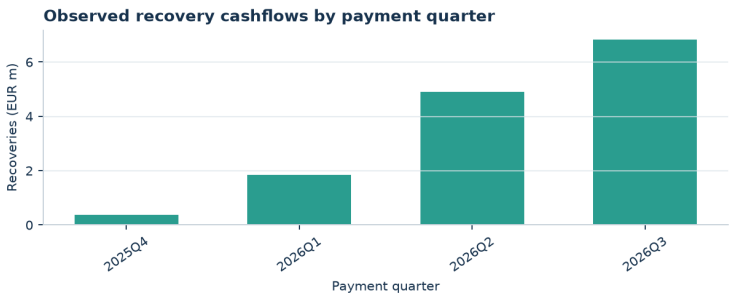
4.2 Collateral type mix

Type	Items	Value	Adjusted	Haircut reduction	Share adjusted
OTHER	101	96,428,944	57,857,366	38,571,578	32.22%
REAL_ESTATE	95	83,640,645	66,912,516	16,728,129	37.26%
FINANCIAL_COLLATERAL	43	57,695,555	54,810,777	2,884,778	30.52%

4.3 Recovery cashflow timing

Payment period	Cashflows	Recovered	Share	Cumulative share
2025Q4	2	371,473	2.66%	2.66%
2026Q1	13	1,831,106	13.13%	15.79%
2026Q2	20	4,909,136	35.20%	51.00%
2026Q3	27	6,833,649	49.00%	100.00%



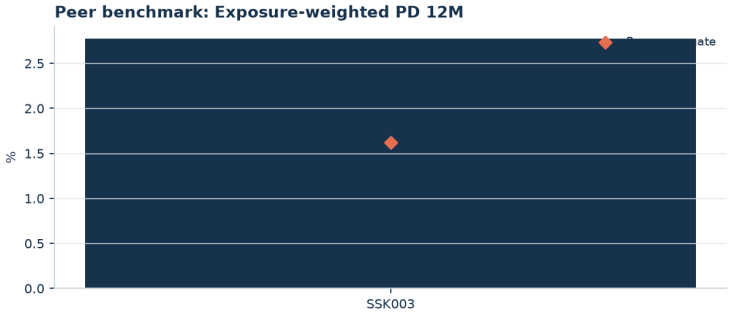


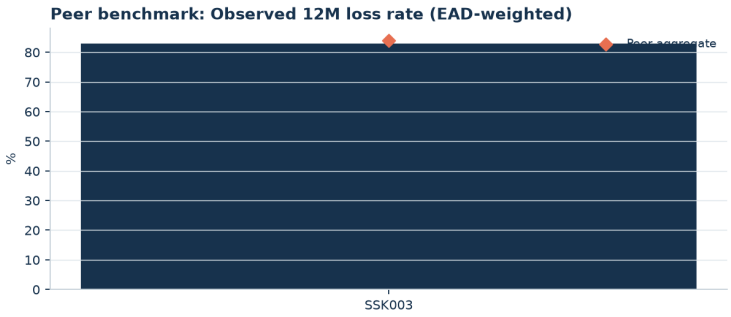
5. Peer benchmarking

Numerator/denominator peer aggregation, excluding the institution from its own peer reference.

Peer values are calculated from summed numerators and denominators, not by averaging bank-level ratios. The bank under review is excluded from its own peer group. Relative differences above 10% are labelled BETTER or WORSE when the metric has an unambiguous direction; neutral rating-share metrics are informational.

Metric	Bank	Peer aggregate	Gap	Rank	Flag
Adjusted collateral coverage of current EAD	22.46%	37.25%	-14.79%	1	WORSE
Observed 12M default rate (count)	10.34%	3.93%	6.41%	1	WORSE
Observed 12M loss rate (EAD-weighted)	82.93%	83.94%	-1.01%	2	IN_LINE
Exposure-weighted PD 12M	2.77%	1.62%	1.15%	1	WORSE
Observed 12M recovery rate (EAD-weighted)	17.07%	16.06%	1.01%	2	IN_LINE

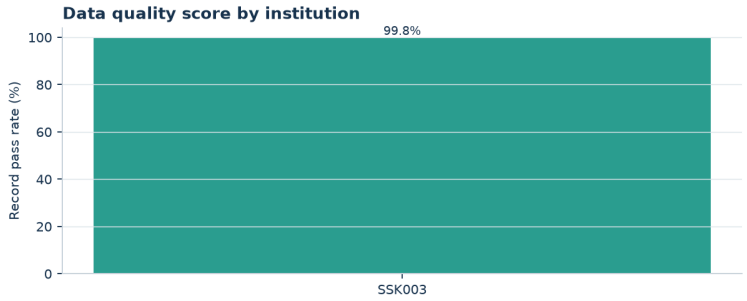




6. Data quality and lineage

Validated records, error quarantine, warnings and pipeline traceability.

The data quality score is 99.82% (valid records / received records). 4 batches completed and 0 duplicate batches were skipped. 19 error findings and 1 warning findings are available in the audit tables. Quarantine holds 17 records; these were not loaded into curated analytical tables. Warnings are retained in curated data with their rule evidence.



6.1 Quality by institution

Bank	Batches	Received	Valid	Rejected	Score	Errors	Warnings
SSK003	4	9,624	9,607	17	9982.00%	19	1

6.2 Top data-quality rules

Rule	Name	Severity	Findings	Records
PLAUS-REC-003	Recoveries do not exceed exposure at default	ERROR	2	2
COMP-EXP-002	Bank ID is required	ERROR	1	1
COMP-EXP-004	Exposure at default is required	ERROR	1	1
COMP-EXP-005	Rating grade is required	ERROR	1	1
CONS-COL-003	Collateral exposure exists	ERROR	1	1
CONS-EXP-002	Rating grade is on the configured scale	ERROR	1	1
CONS-RAT-002	Rating history grade is valid	ERROR	1	1
CONS-RAT-003	Rating history exposure exists	ERROR	1	1
PLAUS-COL-001	Collateral value is non-negative	ERROR	1	1
PLAUS-EXP-001	EAD is non-negative	ERROR	1	1
PLAUS-EXP-006	Reporting date is not in the future	ERROR	1	1
PLAUS-REC-001	Recovery amount is non-negative	ERROR	1	1

6.3 Data lineage and batch audit

Bank	Dataset	Batches	Received	Valid	Rejected	Latest import
SSK003	collateral	1	547	544	3	2026-11-11
SSK003	exposures	1	4,386	4,379	7	2026-11-11
SSK003	rating_history	1	4,387	4,384	3	2026-11-11
SSK003	recovery_cashflows	1	304	300	4	2026-11-11

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7. Methodology and assumptions

Definitions that are deliberately explicit because the prototype is a demonstration.

7.1 Shared data platform

Heterogeneous CSV, Excel and source-database files are mapped into canonical staging entities. Completeness, uniqueness, consistency and plausibility rules run before curated loading. Error records are quarantined with their rule IDs and raw JSON; warnings are retained in curated data. Every batch, quality finding and pipeline run is traceable to a source file hash and import timestamp.

7.2 Rating & Scoring

Rating distribution and expected default are based on the configured, explicitly fictional rating-to-PD mapping: A 0.1%, B 0.3%, C 0.8%, D 2.0%, E 5.0%, F 12.0%. 12-month through-the-cycle-style demonstration PD by rating grade. Values are configurable assumptions, not validated model parameters. Observed default rates use a separate 12-month backward cohort and are not used to calibrate the configured PD in this prototype. Migration compares two snapshots, reports entries and exits separately and uses matched exposures for the transition matrix.

7.3 Loss Estimation

EAD at default is the latest available exposure snapshot on or before the observed default date. Recoveries are cashflows paid after default and on or before the reporting date. Loss is EAD at default minus realised recoveries. No discounting, indirect workout costs or collateral double counting are applied. Collateral coverage uses the latest valuation on or before the reporting date and configured haircuts REAL_ESTATE 20%, FINANCIAL_COLLATERAL 5%, OTHER 40%. Coverage is shown for context, not subtracted again from loss.

7.4 Benchmarking

Benchmark values are calculated from summed numerators and denominators across peer institutions. The institution under review is excluded from its own peer reference. Configured peer groups: PG_URBAN = SSK001, SSK003, SSK005; PG_REGIONAL = SSK002, SSK004. A small peer group means the peer reference is illustrative rather than statistically robust.

7.5 Data quality score

The quality score is the record pass rate: 100 x valid records / received records across completed batches. It is not a measure of model accuracy, calibration or regulatory fitness. The score is computed per batch and aggregated by totals, not by averaging batch percentages.

7.6 Limitations

This is a local DuckDB prototype with synthetic data. It has no real external integrations, no production identity management, no regulatory sign-off and no validated credit-risk models. Peer groups are tiny and illustrative. The configured PDs, haircuts and loss assumptions are demonstration parameters only. A production implementation would require model validation, data-lineage controls, access management, reconciliation, backtesting, audit sign-off and an enterprise database/BI deployment.

7.7 Reproducibility

Synthetic data is generated with a fixed seed and a generation manifest. The pipeline can be rerun with python synthetic_data/generate_data.py followed by python -m etl.pipeline --reset-db. Reports are logged in the report_log table and distribution is simulated locally.

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Appendix A. Report audit

Generated artefact metadata and disclaimer.

Item	Value
Report title	Risk Portfolio Reporting - 2026-09-30
Organisation label	Fictional Sparkassen Reporting Community
Scope	Muster-Sparkasse West (fiktiv)
Peer group	PG_URBAN
Generated at	2026-11-11 00:57:17
Pipeline run	RUN-20261111005418-7B456F2D
Batches completed	20
Rows received	36003
Rows valid	35924
Rows rejected	79
Disclaimer	SYNTHETIC DEMONSTRATION DATA - NOT REGULATORY REPORTING AND NOT REAL SPARKASSEN OR CUSTOMER INFORMATION

SYNTHETIC DEMONSTRATION DATA - NOT REGULATORY REPORTING AND NOT REAL SPARKASSEN OR CUSTOMER INFORMATION

The report is generated from synthetic demonstration data. It must not be represented as a real Sparkasse report, a regulatory submission, a validated model output or a customer communication.